



Course: Data Analysis for Business II Lab			
Course code:	GL2009	Year/Semester:	
Program:	BFT	Units/Cr Hrs:	1
Department:	FAST School of Management	Instructor:	Sadia Manzoor
Course Type:	Core	Email:	sadiamanzoormalik@lhr.nu.edu.pk
Pre-Requisite(s):	None	Phone:	
Prepared By:	Cluster Coordinator (IT & Analytics)	Consultation Hours:	
Approved By:	Head of Department	Approval Date:	15 th July 2024

COURSE DESCRIPTION

The purpose of Data Analysis II Lab is to teach students how to use Statistical software's like SPSS to analyze their data and extract valuable information from the raw data available. Students will be demonstrated on the basic concepts regarding data entry, types of variables, coding of data, hypothesis testing, and various tests like regression and correlation. They will be taught how to use the raw data and arrange it in an organized manner to extract valuable information which would be helpful for the business as a whole.

COURSE OBJECTIVES

Upon successful completion of the course students should be able to:		Program Educational Objectives (PEOs)
1	Gain proficiency in entering, organizing, and presenting data using SPSS.	2
2	Develop skills to create and interpret graphs, charts, and tables for data analysis.	2,3
3	Learn to identify and apply appropriate statistical tests and interpret results.	1,2,4

COURSE LEARNING OUTCOMES (CLOs)

CLO#	Learning Outcome Statement	Program Learning Outcomes (PLOs)	Bloom's Taxonomy Level (BT)
CLO1	Should be able to understand how to enter data, meaning of entered data and variables and Present data in an organized form.	3,7	Understand, Apply
CLO2	Should be able to perform data analysis in forms of graphs, charts, tables and understanding SPSS outputs.	3,4,7	Understand, Apply
CLO3	Should be able to do quantitative data analysis and explain how variables are related and how they impact one another.	3,4,7	Understand, Apply
CLO4	Should be able to decide the type of test suitable for data analysis and interpret the results from the data analysis.	3,4,7	Understand, Apply

LEARNING OUTCOME ASSESSMENT STRATEGY

Assessment Tool	LO1	LO2	LO3	LO4
Quizzes	20	30	20	0
Lab work/ Assignments	15	20	10	30
Mid Term	30	20	0	20
Final Project/ Exam	35	30	70	50

* Percentages are meant to be viewed vertically.



COURSE CONTENTS

CLOs / Week	Contents
CLO1 Week 1	Introduction to the course and SPSS. Discussion of the user interface of SPSS Data Entry Defining Variables
CLO1 Week 2	Editing and coding of data Assigning values to variables Introduction to recoding (same and different variables)
CLO1, CLO2 Week 3	Data analysis (Descriptive, Exploratory and Explanatory) Descriptive Statistics Frequency tables Measures of central tendency (Mean, Median, Mode)
CLO2 Week 4	Compute Function (Calculating sum and means, averages using different functions) Charts, histograms, line graphs, box plots, stem and leaf
CLO2, CLO3 Week 5	Sampling in SPSS Simple Random sampling Split File and Data Analysis
CLO2, CLO3 Week 6	Creating crosstabs Automatic Recoding Handling missing values Quiz 1
CLO4 Week 7	Data distributions Normality tests in SPSS: Q-Q Plots, KS-SW tests, normality curves, stem and leaf, z values etc.
Week 8	Mid Term Exam
CLO3, CLO4 Week 9	Calculating and interpreting confidence intervals Hypothesis Testing Standard hypothesis testing procedure Parametric and non-parametric tests in SPSS
CLO3, CLO4 Week 10	Binomial distribution Introduction to T-tests Introduction to chi square tests Chi square test of goodness of fit
CLO3, CLO4 Week 11	Comparing means through one sample, independent sample t-test and One-Way ANOVA Quiz 2 Conducting and interpreting paired sample t-tests
CLO3, CLO4 Week 12	Correlations using SPSS Simple linear regression & Multiple regression, Logistic Regression
Week 13 & 14	Project Evaluation / Assignment
Week 15	Final Exam

ADMINISTRATIVE INSTRUCTIONS

ASSESSMENT & GRADING

Grading Mechanism	Weightage (%)	Consideration
Project/Final	40%	
Mid Term	20%	1 Mid Term
*Lab Work /Assignments	30%	
*Quiz	5%	
*CP	5%	
Total	100%	

*These are tentative & will be finalized as the semester progresses.

- Grade will be assigned based on the score achieved out of 100%.
- Those scoring below 50% will get F grade.

Course Outline

FAST School of Management

National University of Computer & Emerging Sciences, Lahore Campus



- Please take class participation seriously.

Class Policies

- Please note that any exception for one student is unfair to all other students, so don't expect any.
- Please turn off and store away cell phones, Ipads, laptops, and other electronic devices.
- Talking during lecture is not permitted. It is disrespectful and disruptive to other class members and the instructor.
- If you miss a class, it is your responsibility to determine what was covered including any administrative announcements.

Studying

The proper way of studying for this class is following ADA policy; a short description of the same is as follows

- Ahead of the class, it is expected that you have read the relevant chapters from the textbook;
- During the class you are expected to follow the lecture, take notes and ask questions; and
- After every class you would review your notes and solve the end of chapter exercises and read the textbook.

An extensive set of practice problems is placed on the Xeon Server, make sure you cover them all. We might set a tutorial session from time to time to answer your queries about the problems/practice sets.

Attendance Policy

Students are expected to maintain 100% attendance and ensure a professional demeanor in the class. Active participation in class discussion and activities is highly desirable. Attendance will be marked in the beginning of the class and habitual latecomers will be marked absent. Only under unavoidable circumstances, student can miss a class but minimum 80% attendance is compulsory to sit in final exams as per university policy.

Attendance as well as grades will be regularly uploaded on flex. Students are expected to keep track of their flex records. In case of any discrepancy, let the instructor know at the earliest

Scholastic Dishonesty

In the classroom and in all other academic activities, students are expected to uphold the highest standards of academic integrity. Any form of scholastic dishonesty is an affront to the pursuit of knowledge and jeopardizes the quality of the degree awarded to all graduates.

What exactly is plagiarism?

- Plagiarism is a form of cheating.
- Plagiarism is using someone else's ideas or words and saying they are your own.

If you use material from a text and do not acknowledge the source, you are committing plagiarism.

Specifically, these behaviors are often regarded as plagiarism:

- Copying directly from a text, acknowledging the source but pretending that you are paraphrasing.
- Paraphrasing or copying directly from a text without acknowledging the source.
- Copying from another student's assignment with or without the student's knowledge.

The following behaviors are regarded as misconduct:

- Submitting the same assignment in two different papers.
- Getting someone else to write an assignment for you.

You are also involved in misconduct if you:

- Let another student copy from your own work.
- Write an assignment for another student.

Students who violate the rules on scholastic dishonesty are subject to disciplinary penalties, including the possibility of failure in the course and/or dismissal from the University. *Since dishonesty harms the individual, all students, and the integrity of the University, policies on scholastic dishonesty will be strictly and actively enforced.*